MIDDLE EAST TECHNICAL UNIVERSITY
NORTHERN CYPRUS CAMPUS

DEPARTMENT
OF
COMPUTER ENGINEERING

CNG 351
Data Management and File Structures
Assignment 3

(6% of actual grade)

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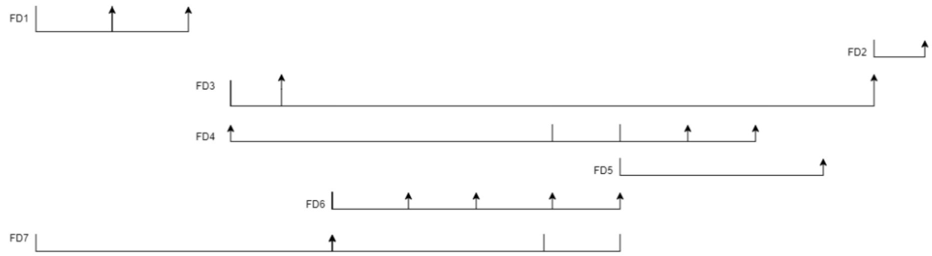
PART1: Normalisation

Case1:

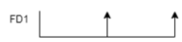
student ID, student Phone,student university, tutor ID, tutor Rating, payment ID, payment Amount, payment Date, lecture Start time,course ID, lecture End Time, lecture Cost, course Name,admin ID, admin Name



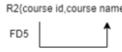
Already in 1NF: no composite or multi-valued attiributes

FD1, FD4, FD5 violates 2NF so decomposed:

R1{student id,student phone, student university}



R2{course id,course name}



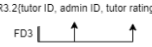
R3{lecture start time, course id,lecture cost, lecture end time, tutor ID, admin ID, admin Name, tutor rating }

R4{student id, lecture start time, course id, payment id,payment Amount, payment Date}

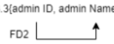
FD2, FD3 and FD6 violates 3NF

R3.1{lecture start time, course id,lecture cost, lecture end time, tutor ID}

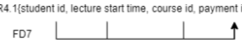
R3.2{tutor ID, admin ID, tutor rating}



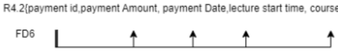
R3.3{admin ID, admin Name}



R4.1{student id, lecture start time, course id, payment id}



R4.2{payment id,payment Amount, payment Date,lecture start time, course id,}



Rest are same

Final Tables

Student(student ID, student Phone, student University)

Course(course ID, course Name)

Lecture(lecture Start Time, course ID, lecture Cost, lecture End Time, tutor ID)

Tutor(tutor ID, tutor Rating, admin ID)

Admin(admin ID, admin Name)

Enroll(student ID[FK:Student:student ID], lecture Start Time[FK:Lecture:lecture Start Time], course ID[FK:Course: course ID], payment ID)

Payment(payment ID, payment Amount, payment Date, lecture Start Time, course Id)

Case2:

playerid	nickname	e score	car name	car brand	car model	raceID	laps	map	position	t id	size	t winner	c id	rarity
2981	snakeserdar	1321	lokmata	Anadol	A5	216	1	Alpine Ascent	5	103	6	5252	20218	Common
2981	snakeserdar	1321	lokmata	Anadol	A5	573	2	Tokyo Driftways	1	101	5	2981	20218	Common
1351	TylerDurden	1169	SoapMachine	Chevrolet	Camaro	848	2	Monaco Circuit	2	101	5	2981	49205	Uncommon
1351	TylerDurden	1169	SoapMachine	Chevrolet	Camaro	526	3	Arctic Edge	3	105	7	2424	49205	Uncommon
1351	TylerDurden	1169	doNotT4lk	Suzuki	Swift RS	482	3	Istanbul Inter	2	105	7	2424	12151	Uncommon
5252	kharphaz	959	McQueen	Chevrolet	Corvette	216	1	Alpine Ascent	1	103	6	5252	14300	Rare
5252	kharphaz	959	scubaRacer	BMW	1.16	482	3	Istanbul Inter	4	105	7	2424	56980	Common
3459	LadyDeath	803	McQueen	Subaru	BRZ	526	3	Arctic Edge	7	105	7	2424	59143	Uncommon
2424	DominicToretto	1520	FamilyFirst	Dodge	Charger	431	2	Sahara Speedway	2	105	7	2424	83906	Common
2424	DominicToretto	1520	FamilyFirst	Dodge	Charger	526	3	Arctic Edge	1	105	7	2424	83906	Common
3153	SametBaba	1046	kolpha	Tesla	Roadster	216	1	Alpine Ascent	3	103	6	5252	71338	Rare
3153	SametBaba	1046	kolpha	Tesla	Roadster	106	3	Rio Rally	1	103	6	5252	71338	Rare
3153	SametBaba	1046	DocHudson	Hudson	Hornet	482	3	Istanbul Inter	3	105	7	2424	31263	Common
7045	xKralTR	1091	Kha-Rhambhol	Audi	R8	848	2	Monaco Circuit	4	101	5	2981	97072	Rare
4019	yesilady	1167	Cardinality	Mazda	Miata	106	3	Rio Rally	2	103	6	5252	48190	Common
4019	yesilady	1167	Cardinality	Mazda	Miata	681	3	Death Valley Run	1	102	5	1351	48190	Common
2147	Sp33d4dd1ct	1428	GottaGoF4st	Bugatti	Tourbillon	431	2	Sahara Speedway	8	102	5	1351	56897	Common
PK														
FK			Composite Key											
						PK								
										PK		FK		
													PK	
						PK				FK				
FK						Composite								

Dependencies for given table:

- FD1 :{playerid}– >{nickname, e score}
- FD2 :{playerid(fk),car name}– >{car brand, car model}
- FD3 :{raceID}– >{laps, map}
- FD4 :{t id}– >{size, t winner(fk)}
- FD5 :{c id}– >{rarity}
- FD6 :{race id}– >{t id(fk)}
- FD7 :{race id,playerid(fk)}– >{position}

After normalized to 3NF:

For normalizing process to 3NF I split relation by just dependencies to primary keys

{playerid}– >{nickname, e score} becomes {playerid}– >{nickname} and { nickname }– >{e score}

player id	nickname
2981	snakeserdar
1351	TylerDurden
5252	kharphaz
3459	LadyDeath
2424	DominicToretto
3153	SametBaba
7045	xKralTR
4019	yesilady
2147	Sp33d4dd1ct

nickname	e score
snakeserdar	1321
TylerDurden	1169
kharphaz	959
LadyDeath	803
DominicToretto	1520
SametBaba	1046
xKralTR	1091
yesilady	1167
Sp33d4dd1ct	1428

{playerid(fk),car name}– >{car brand, car model}

player id	car name	car brand	car model
2981	lokmata	Anadol	A5
1351	SoapMachine	Chevrolet	Camaro
1351	doNotT4lk	Suzuki	Swift RS
5252	McQueen	Chevrolet	Corvette
5252	scubaRacer	BMW	1.16
3459	McQueen	Subaru	BRZ
2424	FamilyFirst	Dodge	Charger
3153	kolpha	Tesla	Roadster
3153	DocHudson	Hudson	Hornet
7045	Kha-Rhambhol	Audi	R8
4019	Cardinality	Mazda	Miata
2147	GottaGoF4st	Bugatti	Tourbillon

{raceID}– >{laps, map}

raceID	laps	map
216	1	Alpine Ascent
573	2	Tokyo Driftways
848	2	Monaco Circuit
526	3	Arctic Edge
482	3	Istanbul Inter
431	2	Sahara Speedway
106	3	Rio Rally
482	3	Istanbul Inter
681	3	Death Valley Run

{t id}– >{size, t winner(fk)}

t id	size	t winner
103	6	5252
101	5	2981
105	7	2424
102	5	1351

{c id}– >{rarity}

c id	rarity
20218	Common
49205	Uncommon
12151	Uncommon
14300	Rare
56980	Common
59143	Uncommon
83906	Common
71338	Rare
31263	Common
97072	Rare
48190	Common

{race id}– >{t id(fk)}

raceID	t id
216	103
573	101
848	101
526	105
482	105
482	105
431	105
106	103
681	102

{race id,playerid(fk)}– >{position}

raceID	player id	position
216	2981	5
573	2981	1
848	1351	2
526	1351	3
482	1351	2
216	5252	1
482	5252	4
526	3459	7
431	2424	2
526	2424	1
216	3153	3
106	3153	1
482	3153	3
848	7045	4
106	4019	2
681	4019	1
431	2147	8

After normalized to BCNF:

{raceID}– >{laps, map} becomes {raceID}– >{laps} and { map }– >{ laps } because violates BCNF

raceID	laps
216	1
573	2
848	2
526	3
482	3
431	2
106	3
482	3
681	3

map	laps
Alpine Ascent	1
Tokyo Driftways	2
Monaco Circuit	2
Arctic Edge	3
Istanbul Inter	3
Sahara Speedway	2
Rio Rally	3
Istanbul Inter	3
Death Valley Run	3

Rest of the tables are same.

PART2: Relational algebra

1. π_model,os(Device)
2. σ_status='approved' & (timestamp > '2024-11-11') ^ (π_deviceMacAddress^(Registration))
3. σ_type='laptop' ^(Device) ⋈ Device.owner = Student.studentID^(π_name,surname^(Student))
4. σ_status='declined' & timestamp >= '2024-01-01' ^(Registration)⋈ Registration.deviceMacAddress = Device.macAddress⋈ Device.owner = Student.studentID (π_name,surname^(Student))
5. (σ_name='Ahmed' ∧ surname='Zorlu' (Student)⋈ Student.studentID = Device.owner⋈ Device.macAddress = Registration.deviceMacAddress⋈ Registration.regID = BlockHistory.regID⋈ BlockHistory. blockedBy = NetworkEngineer.engID (σ_name='Mehmet' ∧ surname='Muhendis' (NetworkEngineer)) (π_regID,startTimestamp,duration,description^(BlockHistory))
6. (ρ_R(sum_of_wired) ℑ_SUM duration^(σ_type='wired'^(Registration) ⋈ Registration.regID = BlockHistory.regID^(Block History))) U (ρ_R(sum_of_wireless) ℑ_SUM duration^(σ_type= wireless ^(Registration) ⋈ Registration.regID = BlockHistory.regID^(Block History)))